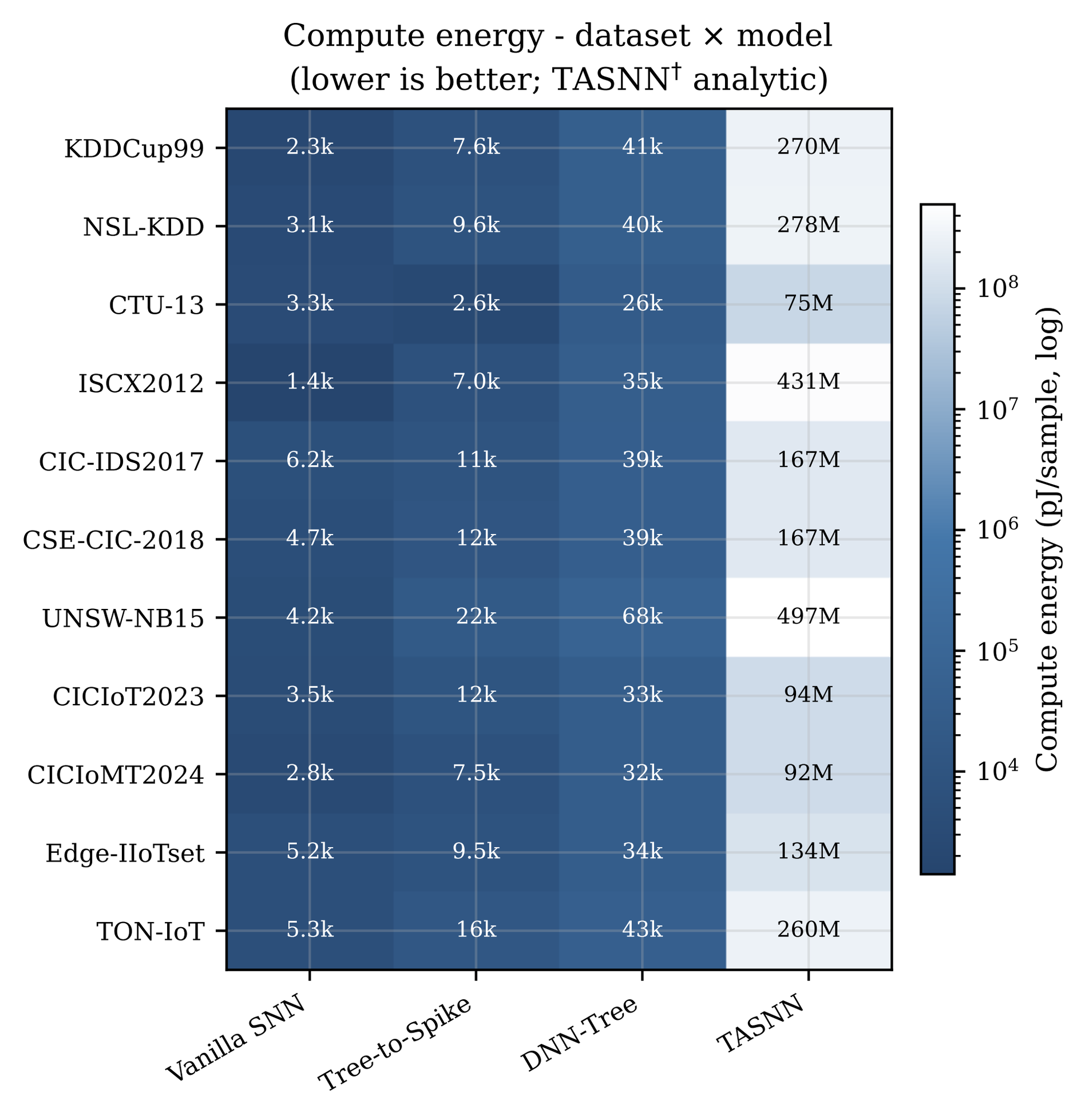
Compute energy - dataset × model
(lower is better; TASNN† analytic)
KDDCup99
NSL-KDD
CTU-13
ISCX2012
CIC-IDS2017
CSE-CIC-2018
UNSW-NB15
CICIoT2023
CICIoMT2024
Edge-IIoTset
TON-IoT
2.3k
7.6k
41k
270M
3.1k
9.6k
40k
278M
3.3k
2.6k
26k
75M
1.4k
7.0k
35k
431M
6.2k
11k
39k
167M
4.7k
12k
39k
167M
4.2k
22k
68k
497M
3.5k
12k
33k
94M
2.8k
7.5k
32k
92M
5.2k
9.5k
34k
134M
5.3k
16k
43k
260M
Vanilla SNN
Tree-to-Spike
DNN-Tree
TASNN
Compute energy (pJ/sample, log)
10^8
10^7
10^6
10^5
10^4